Project 2: Loan Repayment Prediction

Machine Learning Classification Project

1. Objective

The objective is to predict whether a loan will be paid back using applicant financial and categorical information. Four classification models were developed and compared: Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM).

2. Dataset & Preparation

The dataset contains 593,994 records and 13 original columns. The target variable is **loan_paid_back**, where 1 represents a paid-back loan and 0 represents a loan not paid back. There were no missing values, duplicate rows, or duplicate IDs. The ID column was removed before modeling.

The predictors included annual income, debt-to-income ratio, credit score, loan amount, interest rate, gender, marital status, education level, employment status, loan purpose, and grade/subgrade. Numerical features were standardized and categorical features were one-hot encoded. An 80:20 stratified split produced 475,195 training records and 118,799 test records.

3. Exploratory Data Analysis

The target was imbalanced: 79.88% of observations were class 1 and 20.12% were class 0. Gender showed only small differences in observed repayment rates. Employment status showed a much stronger association: Retired 99.72%, Self-employed 89.85%, Employed 89.41%, Student 26.35%, and Unemployed 7.76%.

Loan purpose and education level showed more moderate differences. Correlation analysis found debt-to-income ratio had the strongest negative linear association with repayment (-0.34), while credit score had a positive association (+0.23). Interest rate showed a weaker negative association (-0.13). These are associations, not causal conclusions.

4. Model Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	86.14%	93.83%	88.47%	91.07%	N/A
Decision Tree	86.17%	94.10%	88.22%	91.06%	91.12%
Random Forest	90.09%	90.23%	98.23%	94.06%	90.70%
SVM	86.64%	93.63%	89.35%	91.44%	91.05%

Random Forest produced the strongest overall results: 90.09% accuracy, 98.23% recall, and the highest F1 score of 94.06%. Its confusion matrix was [[13,811, 10,089], [1,681, 93,218]].

For computational practicality, SVM was trained using a stratified 100,000-row sample of the training set. Evaluation used the same 118,799-row test set. A linear SVM was used after full-scale kernel SVM with probability estimation proved computationally impractical.

5. Best Model & Business Interpretation

Random Forest was selected as the best overall model. It achieved the highest accuracy, recall, and F1 score. It can be used as a decision-support tool to estimate loan repayment outcomes and flag applications for additional review. Final lending decisions should still involve appropriate human and policy review.

6. Conclusion

The project successfully completed data preparation, EDA, classification modeling, evaluation, and model comparison. Random Forest provided the strongest overall predictive performance for this dataset. Future work could include hyperparameter tuning, threshold optimization, probability calibration, fairness checks, and validation on future data.